

Kamal Joshi

Senior Software Engineer — Cloud Gateway & API Infrastructure | Go | Java | Kubernetes | Distributed Systems | Observability

Austin, TX | Open to remote-friendly and hybrid roles

+1-614-825-3047 | kamal2311@gmail.com | linkedin.com/in/kamaljoshiworks | github.com/kamal2311

PROFESSIONAL SUMMARY

Senior software engineer with deep hands-on experience designing, building, and operating API gateway infrastructure and distributed backend services in cloud environments — including production ownership of a Kong-based API gateway and an integrations platform that today powers Cisco's integrations with Google, Microsoft, and Amazon. Proven track record of end-to-end ownership: from architecture decisions and rollout planning through incident response and systemic reliability improvements. Strong at breaking down complex, multi-service problems using data, driving stakeholder alignment, and mentoring engineers through detailed design and code reviews. Effective at directing AI tools (Claude Code, GitHub Copilot, Codex) across the full SDLC to ship high-quality work faster.

CORE STRENGTHS

- **API Gateway & networking:** Kong API Gateway (production ownership), gRPC, gRPC-Web, HTTP/S, REST, L7 API design, API key management, rate limiting, OAuth 2.0 / OIDC token flows, TLS secret management
- **Distributed systems:** Go, Java, microservices, event-driven architecture, SQS/SNS, DynamoDB Streams, CloudEvents, high-throughput message delivery, fault tolerance, consistency tradeoffs
- **Cloud & containers:** AWS (Lambda, ECS, EKS, S3, SQS/SNS, DynamoDB, Secrets Manager), Kubernetes, Docker, multi-region deployments, sovereign/GovCloud environments
- **Observability & operational excellence:** metrics instrumentation, alerting, DLQ monitoring, incident response, retention-risk alerting, rate-limit governance, deployment validation
- **Infrastructure as Code & CI/CD:** Terraform modules, Terraform pipelines, multi-account infrastructure, automated rollouts, serverless CI/CD pipelines
- **Technical leadership:** architecture design, project ownership, cross-team coordination, mentoring, design reviews, pragmatic tradeoff decisions
- **AI-assisted SDLC:** Claude Code, GitHub Copilot, Codex — daily use across implementation, IaC, testing, debugging, and code review; pioneered team-wide AI adoption

EXPERIENCE

Cisco

Software Engineering Technical Leader

November 2025 - Present | Austin, TX

- Owned production design and operation of **Kong API Gateway**, **Authorization Service**, and **Third-Party Integrations Service** — the API infrastructure backbone for Cisco Secure Access serving enterprise and government customers across multiple regions and clouds.
- Founded the Third-Party Integrations Service from scratch: designed a unified API scheme extensible enough to accommodate Chrome Enterprise, Microsoft Intune, Jamf Pro, Splunk, and CloudEvents integrations without structural rewrites; the service now powers Cisco's integrations with **Google, Microsoft, and Amazon** and enabled deals at major enterprise accounts.
- Mapped **AuthSvc TLS secret synchronization and API key refresh behavior** across Secrets Manager, Kubernetes, and Kong — reducing ambiguity for security-sensitive gateway automation and token lifecycle management.
- Prevented security-event message loss during a production incident by identifying a **300k+ message DLQ backlog**, quantifying impact across **15 orgs**, and recovering within **5-10 minutes** — then drove systemic observability and retention-risk alerting improvements.

- Resolved a customer-scale onboarding failure by tracing **250+ appliance registration requests** through API key authentication and Kong rate-limit behavior, unblocking the customer and driving durable fixes around retry/backoff, token reuse, and service ownership.
- Drove multi-region production readiness by decomposing a high-risk **Terraform** rollout into staged changes across ECR, Secrets Manager, DynamoDB Streams, SNS, and service config — validating HA behavior across multiple regions.
- Reframed API onboarding as a control-plane problem and proposed a **hybrid Kong Admin API + Terraform model** — a platform improvement that increased self-service speed while preserving auditability.

Senior Software Engineer

January 2021 - November 2025 | Austin, TX

- Implemented **metrics instrumentation** for the API Gateway and Integrations Service; owned production observability, alerting, and DLQ monitoring for gateway-adjacent backend services.
- Built **auth-tools**, an internal authorization library adopted across multiple platform services to simplify OAuth 2.0 / policy enforcement integration — a reusable component that reduced integration friction team-wide.
- Mentored early-in-career engineers through code review and design feedback; initiated **“Tip Talk Tue”**, a recurring knowledge-sharing forum on engineering productivity and best practices.
- Integrated AI tooling (Claude Code, Copilot, Codex) into the full SDLC; **pioneered team-wide GitHub Copilot adoption** ahead of broader Cisco initiatives.

Randstad Technologies US at Cisco

Senior Software Developer

March 2015 - January 2021 | Dublin, OH

- Designed and implemented **Go-based gRPC and HTTP/S APIs** for a cloud-native data ingestion platform.
- Built API-driven **OIDC and OAuth 2.0** authentication and authorization flows using gRPC-Web interactions.
- Contributed to automatic generation of **protocol buffer specs and gRPC APIs** from SQL schemas, improving API consistency across services.
- Reduced **100+ OWASP-reported security warnings to zero** through secure coding, browser security headers, input sanitization, and server-side validation.

Earlier Engineering Experience

Cognizant, Oracle, Collabera | December 2007 - February 2015

- Delivered backend services, REST APIs, and integration features across financial services, healthcare, insurance, and public-sector domains.
- Built a foundation in Java, REST APIs, SQL, distributed service patterns, and maintainable engineering practices.

CERTIFICATIONS

- AWS Certified Solutions Architect - Associate
- AWS Certified Developer - Associate
- MongoDB Certified Java Developer M101J
- Google Cloud Platform Essentials

EDUCATION

Bachelor of Engineering, Information Technology
Dharmsinh Desai University